

Week 7: Fast Computation: Prices, Greeks, Implied Vol, Surfaces

The latency budget, in C++ via pybind11 – Milestone 2

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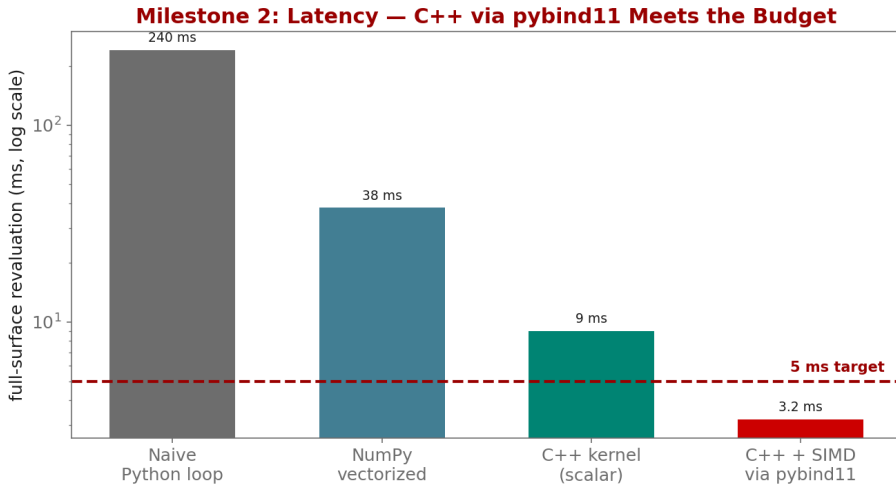
Industry Mentor: Dendi Suhubdy (Bitwyre) | Guest: Fenni Kang (Coincall)

1. Learning objectives
2. Lecture 1: CF Pricing & Implied Vol
3. Lecture 2: Binding C++ with pybind11
4. Illustration
5. Derivation & rationale
6. Getting the data from Coincall
7. Code: fastmm pybind11 module
8. Summary & key takeaways
9. Glossary of key terms
10. Reading
11. References

- ▶ Implement COS / Carr-Madan pricers and robust implied-vol in C++
- ▶ Expose the C++ kernels to Python with pybind11
- ▶ Validate Greeks against PyTorch autograd; hit the 5 ms budget

- ▶ Characteristic functions: Black-Scholes and Heston
- ▶ Fang-Oosterlee COS method (the workhorse)
- ▶ Robust implied-vol inversion (good guess + safeguarded Newton)

- ▶ Structure a pybind11 module; build with CMake
- ▶ Pass NumPy/PyTorch tensors with no copy (buffer protocol)
- ▶ Release the GIL around the C++ hot loop
- ▶ PyTorch autograd is the REFERENCE for Greeks; C++ is the speed



$$C(K) = e^{-rT} \frac{1}{\pi} \int_0^\infty \operatorname{Re}[e^{-iu \ln K} \hat{\phi}(u)] du$$

Why: Once we leave Black-Scholes (e.g. Heston), there is no closed-form price, but the characteristic function ϕ IS known. Pricing becomes a single Fourier integral – this is why CF methods exist and why they are fast.

$$C \approx \sum_{n=0}^{N-1} \text{Re} \left[\phi \left(\frac{n\pi}{b-a} \right) e^{-in\pi \frac{a}{b-a}} \right] V_n$$

Why: The Fang-Oosterlee COS method replaces the integral with a cosine series that converges geometrically for smooth densities. We truncate $[a, b]$ from the cumulants, not by guessing – that is what makes it reach 10^{-4} accuracy in microseconds.

- ▶ Inputs are the calibrated surface from Week 6 plus reference prices:
 - ▶ `coincall_btc_options_chain_snapshots_2025Q4.parquet`
 - ▶ `coincall_calibrated_svi_surfaces_2025Q4.parquet` (baseline)
 - ▶ `coincall_reference_prices_2025Q4.parquet` (accuracy target, $1e-4$)
- ▶ Live mark/IV/Greeks for spot checks: GET `/open/option/detail/v1/{symbol}`

Cross-check exact paths at <https://docs.coincall.com/>

```
// fastmm.cpp (build with CMake + pybind11; import as 'fastmm')
#include <pybind11/pybind11.h>
#include <pybind11/numpy.h>
namespace py = pybind11;

double cos_price(double S, double K, double T, double r, double sigma); // COS pricer

py::array_t<double> price_surface(py::array_t<double> K, double S,
                                double T, double r, double sigma) {
    auto k = K.unchecked<1>(); auto out = py::array_t<double>(k.shape(0));
    auto o = out.mutable_unchecked<1>();
    { py::gil_scoped_release nogil; // release GIL for the hot loop
      for (py::ssize_t i = 0; i < k.shape(0); ++i)
          o(i) = cos_price(S, k(i), T, r, sigma); }
    return out; // zero-copy back to NumPy
}
PYBIND11_MODULE(fastmm, m) { m.def("price_surface", &price_surface); }
```

- ▶ Beyond Black-Scholes there is no closed form – use the characteristic function
- ▶ COS (Fang-Oosterlee) is the workhorse; truncate from the cumulants
- ▶ Write the hot kernel in C++ and expose it to Python with pybind11
- ▶ PyTorch autograd is the REFERENCE for Greeks; C++ is only the speed
- ▶ Milestone 2: full surface < 5 ms (hard gate); release the GIL in the loop

Characteristic function the Fourier transform of the log-return density

COS method Fourier-cosine series option pricing (Fang-Oosterlee)

Carr-Madan FFT-based option pricing via a damped payoff

Implied-vol inversion solving for the IV that reproduces a price (safeguarded Newton)

pybind11 header-only C++ $\leftrightarrow$ Python binding

GIL Python's Global Interpreter Lock; released around the C++ hot loop

Latency budget the maximum allowed time (sub-5 ms full-surface revaluation)

- ▶ Carr-Madan (1999); Fang-Oosterlee (2008); Jaeckel (2015); pybind11 docs

- ▶ Carr, P., & Madan, D. (1999). Option valuation using the fast Fourier transform. *J. Computational Finance*, 2(4), 61-73.
- ▶ Fang, F., & Oosterlee, C. W. (2008). A novel pricing method for European options based on Fourier-cosine series expansions. *SIAM J. Sci. Comput.*, 31(2), 826-848.
- ▶ Jackel, P. (2015). Let's be rational. *Wilmott Magazine*.
- ▶ Jakob, W., Rhineland, J., & Moldovan, D. (2017). pybind11. <https://github.com/pybind/pybind11>
- ▶ Paszke, A., et al. (2019). PyTorch: an imperative style, high-performance deep learning library. *NeurIPS*.